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Chart Plotting

Visual study guide — diagrams, memory hooks & exam traps



Plotting tools · Latitude, longitude & distance · Courses from the rose · DR & labels
Bearings & fixes · Set & drift · Course to steer · Light ranges · 60D = ST
Includes a fictional practice chart with 8 worked problems
Prepared for Jesse Orebaugh · October 2026

How this exam module works

Chart Plotting is the shortest module, **10 questions**, but it's one of the two that need **90%**: you can miss **one**. It's open book and allows up to 3.5 hours, so speed isn't the issue. Neat, careful plotting is.

Module	Questions	Pass	Misses allowed
Rules of the Road	50	90%	5
Deck General & Safety	50	70%	15
Navigation General	50	70%	15
Chart Plotting	**10**	**90%**	**1**

MLS final format per Mariners Learning System. Confirm the exact chart and tools in your course materials.

What you need on the table

- **The chart from your course** (exam chart plots are done on a NOAA training chart; MLS tells you which).
- **Parallel rulers** (or a Weems-style plotter) to carry directions to and from the compass rose.
- **Dividers** for distance, measured on the latitude scale.
- Sharp pencil, eraser, and a calculator for 60D = ST and the light-range formula.

How to work every question

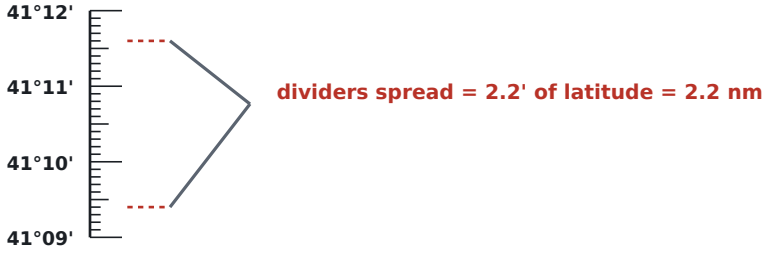
- **Plot everything, label everything.** Course above the line, speed below, time at every position.
- **Work in TRUE on the chart.** Convert compass or magnetic numbers to true before you draw (TVMDC).
- **Check the answer choices for spacing.** If two choices are 1° apart, replot carefully; if they're 10° apart, a rough plot will do.
- **Problems chain.** A later question often starts from an earlier fix. If one answer looks way off, recheck the earlier plot.

MEMORY HOOK **The aviator's edge**

This is VFR flight planning on a sectional: true course off the chart, correct for variation, compute ground speed and ETA, and fix your position by crossing radials. Same skills, slower vehicle.

Reading the chart

Latitude, longitude and distance



Each minute of LATITUDE = 1 nautical mile. Small ticks are tenths (0.1' = 0.1 nm). Measure on the side scale, level with the middle of the leg you're measuring.

Never measure on the top/bottom (longitude) scale: those minutes shrink.

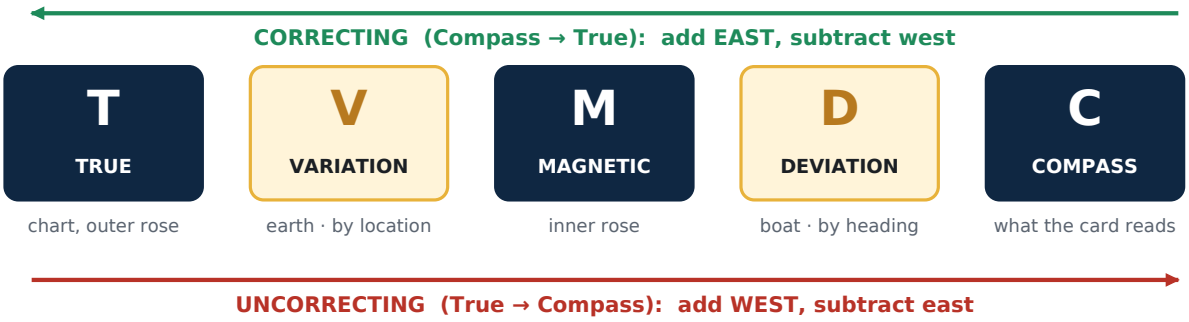
Task	How
Read latitude	Side scales. Degrees, minutes, tenths of a minute: 41°09.0'N
Read longitude	Top and bottom scales. West longitude **increases to the left** on U.S. charts
Plot a position	Lay the parallel rulers along a latitude line, slide to the latitude; step the longitude in with dividers
Measure distance	Span the leg with dividers, then lay them on the **latitude scale beside the leg**
Long legs	Set the dividers to a round number (5 nm) and walk them along the line

EXAM TRAP **Latitude scale only**

Using the longitude scale is the classic chart-plot mistake. At 41°N a minute of longitude is only about 0.75 nm.

Courses from the compass rose

- Lay the parallel rulers along your course line, then walk them to the rose so one edge passes through its **center**.
- Read the **outer ring for true**, the inner ring for magnetic. Read on the side the boat is heading toward.
- Exam answers are usually asked in true. If a question wants magnetic or compass, convert with TVMDC rather than reading the inner ring; it's more precise.



True Virgins Make Dull Companions · Can Dead Men Vote Twice

Dead reckoning and labeling

A **DR** position uses only course and speed from the last fix. It ignores current, which is what makes set and drift measurable later.



○ **circle**
FIX (2+ LOPs)
◠ **half circle**
DR (course & speed only)
□ **square**
EP (DR + best estimate)
Label above line
Course: C 233°T
Label below line
Speed: S 10
Times
4-digit, 24-hour

Speed, time, distance: 60D = ST

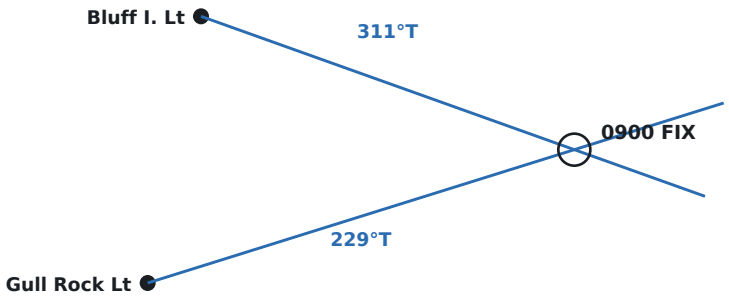
Find	Formula	Example
Distance	D = S × T ÷ 60	10 kn for 45 min = 7.5 nm
Time (minutes)	T = 60 × D ÷ S	13.0 nm at 10 kn = 78 min
Speed	S = 60 × D ÷ T	6.4 nm in 45 min = 8.5 kn

MEMORY HOOK **Sixty D Street**
60 × Distance = Speed × Time, with time in minutes. Cover the one you want and solve for it.

DR rules

- Start a new DR line at every **fix** and every **course or speed change**.
- Plot a DR at least every hour, and at the time of each question.
- Label the DR with its time (0900 DR). Don't move the DR when you get a fix; the gap between them is your current.

Bearings and fixes



1. **Convert each bearing to TRUE** (hand compass: deviation = 0)
2. **Lay parallel rulers on the rose,** walk them to the object
3. **Draw the line toward you (the** reciprocal side of the object)
4. **Where the lines cross = FIX.** Best when they cross near 90°.

Converting the bearing

Bearing taken with	Correction to get true
Hand-bearing compass	Variation only (deviation is 0)
Ship's steering compass	Deviation for the **ship's heading**, then variation
Magnetic bearing given	Variation only
Relative bearing	Add the ship's true heading (subtract 360 if over)

EXAM TRAP **Deviation uses the SHIP'S heading**

Taking a bearing with the steering compass? Look up deviation for the direction the **boat** is heading, not the direction of the bearing.

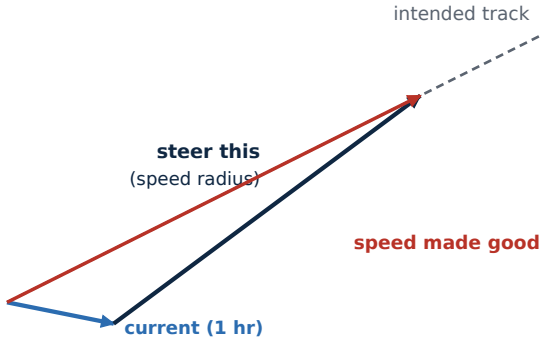
Other position tools

Tool	What it is
Three-bearing fix	The small triangle ("cocked hat") means some error; take the center or the corner nearest danger
Range	Two objects in line: an exact LOP with no compass error at all
Running fix	Advance an earlier LOP along your course by the distance run, cross it with a new LOP on the same or another object
Bearing + distance	One LOP plus a radar or charted distance off gives a fix

Current: set, drift and course to steer



Finding set & drift



Course to steer

Set and drift from a fix

- Plot your DR for the time of the fix.
- **Set** = true direction **from the DR to the fix**.
- **Drift** = distance from DR to fix ÷ hours since the last fix.

Course to steer (current triangle)

- From your start, draw the **intended track** to the destination.
- From the start, draw the **current vector** for one hour (set, length = drift).
- From the end of the current vector, swing the dividers open to your **speed** until they touch the track.
- The line from the current's tip to that point is the **course to steer**. Start to that point is the **speed made good**.

MEMORY HOOK **It's a wind triangle**

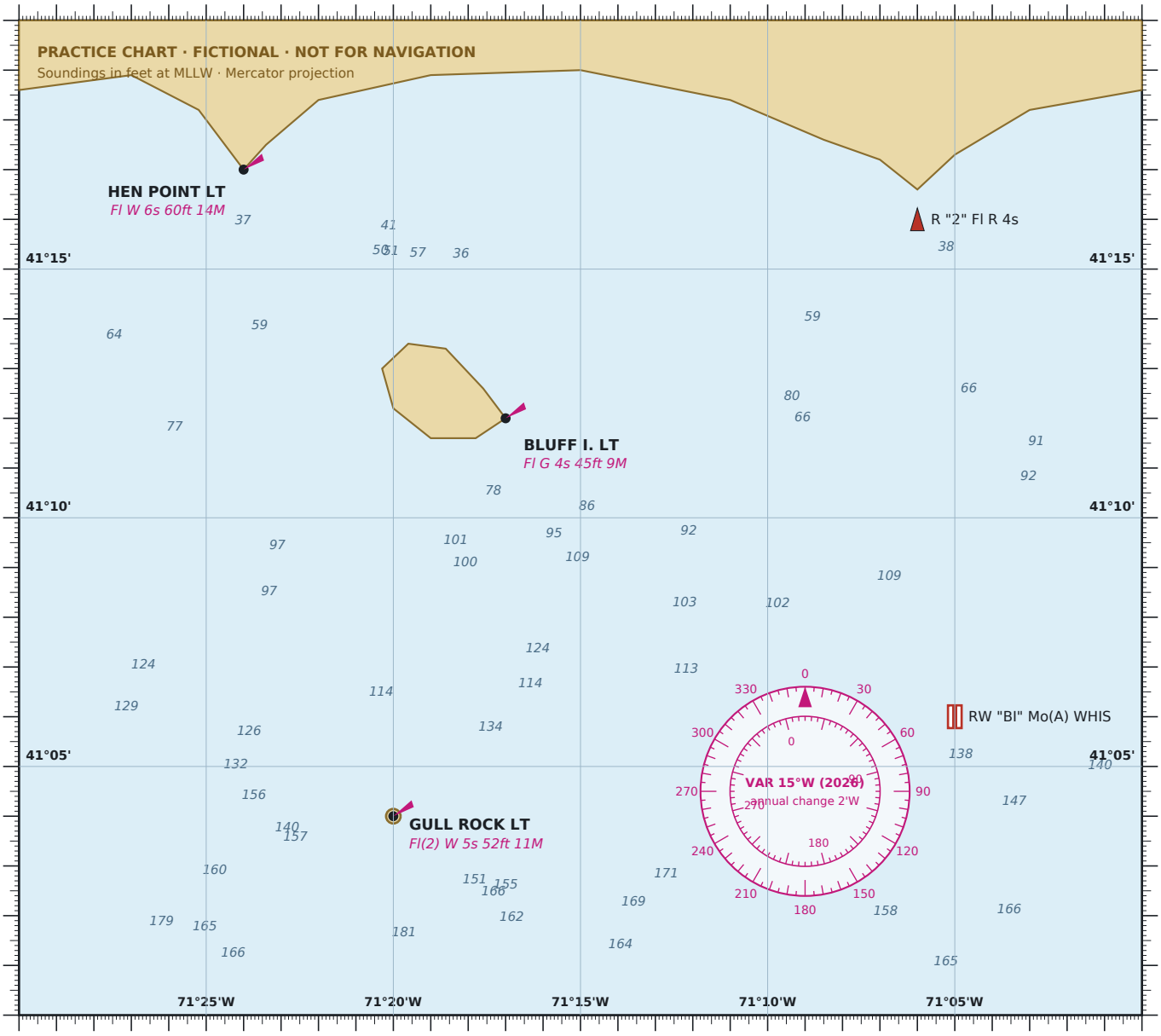
Same as wind correction angle in flight planning: crab into the current, and ground speed comes from the third side.

Lights and danger

Question	How
When will a light rise above the horizon?	Geographic range = 1.17 × (√height of light + √height of eye), feet in, nm out
Will I see it at all?	First sighting is the **lesser** of geographic and luminous range
Danger bearing	A charted bearing to an object that keeps you off a hazard: stay NMT (not more than) or NLT (not less than) the line

Practice chart

A fictional chart drawn to Mercator scale (latitude 41°00'–41°20'N, longitude 71°00'–71°30'W). Print this page and plot the problems on the next page with real tools, or work them on a screen with a ruler app.



Deviation card (by compass heading)

Compass hdg	000°	045°	090°	135°	180°	225°	270°	315°
Deviation	2°W	1°W	0°	1°E	2°E	2°E	1°E	1°W

Interpolate between headings. A hand-bearing compass has no deviation.

Practice problems

1. You depart point **A** at 41°15.0'N 71°08.0'W at **0815**, bound for point **B** at 41°05.0'N 71°19.0'W. What is the **true course** and **distance** from A to B?

2. Variation is 15°W (use the rose). Deviation on this heading is 2°E (see the card). What **compass course** do you steer?

3. At **10 knots**, what is your ETA at B if nothing pushes you off course?

4. Plot your **0900 DR**. What are its latitude and longitude?

5. At 0900 you take hand-bearing compass bearings: Bluff I. Lt **326° magnetic** and Gull Rock Lt **244° magnetic**. What are the true bearings, and what is your **0900 fix**?

6. Using your 0900 DR and fix, what are the **set and drift** of the current since 0815?

7. From the 0900 fix, still making 10 knots through the water and assuming the same current, what **true course do you steer** to reach B, what **speed do you make good**, and what is your new **ETA**?

8. Gull Rock Lt is 52 ft high and your height of eye is 9 ft. How far off will the light first rise above the horizon (geographic range)? Will its 11 nm nominal range limit that on a clear night?

> **TIP** **Tolerances**
>
> By hand, being within **±1°** and **±0.1–0.2 nm** of the answer key means your plot is good. Exam choices are usually farther apart than that.

Answers with the work

1. Course **220°T**, distance **13.0 nm**.

2. T 220 + 15W = M 235 − 2E = **233° compass**.

3. 60 × 13.0 ÷ 10 = 78 min. 0815 + 78 min = **ETA 0933**.

4. 45 min at 10 kn = 7.5 nm along 220°T: **41°09.2'N 71°14.4'W**.

5. Magnetic 326 and 244, minus 15°W = true **311°T** and **229°T**. Fix: **41°09.0'N 71°12.4'W**. The lines cross at about 82°, a strong fix.

6. DR to fix bears **099°T** (set), 1.5 nm in 45 min = **2.0 kn** (drift).

7. Track fix → B is 231°T, 6.4 nm. Current triangle: steer **240°T** (255°M), speed made good **8.5 kn**, 45 min, **ETA 0945**.

8. 1.17 × (√52 + √9) = 1.17 × (7.21 + 3.00) = **11.9 nm**. The 11 nm nominal range is less, so on a clear night the light's brightness limits you first: expect it at about **11 nm** (the lesser of the two).

EXAM TRAP **Notice what the current did**

Ignoring the current (problem 3) gives ETA 0933. Correcting for it (problem 7) gives 0945, and an uncorrected boat would have been set well east of B. Exam chains test exactly this.

Exam-day checklist

- Read the whole problem set first; mark which questions build on earlier positions.
- Write TVMDC on scratch paper; convert every bearing and course to true before you draw.
- Label every line (course above, speed below) and every position with time and symbol.
- Measure distance on the latitude scale beside the leg; recheck any distance by walking the dividers.
- After each answer, sanity check: does the direction look right on the chart? Does the time make sense at that speed?
- Use the time: 3.5 hours for 10 questions. Replot anything where two answer choices are close.

Quick reference

Item	Rule
Correcting (C → T)	Add east, subtract west
Uncorrecting (T → C)	Add west, subtract east
1' of latitude	1 nautical mile
60D = ST	Time in minutes
Set	Direction DR → fix
Drift	DR-to-fix distance ÷ hours
Geographic range	1.17 × (√HoL + √HoE)
Symbols	○ fix · ◠ DR · □ EP

TIP **Get the real chart**

Practice on the exact chart your MLS course names, with the tools you'll use on exam day. This guide's chart teaches the method; the real chart builds speed with its specific lights and scales.